iNTRA Solutions

PSU OJT MANAGEMENT & MONITORING PORTAL

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iNTRA Solutions

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I. Introduction

Mission and Vision

Mission 01

To empower educational institutions through innovative, efficient, and integrated digital solutions that shape the future of academic management.

Vision 02

To deliver secure, user-centered, and sustainable IT systems that promote operational transparency, efficiency, and service excellence in higher education.

Who We Are

iNTRA Solutions

TechnoSphere Solutions is a student-founded technology firm driven by the goal of transforming educational institutions through digital innovation. The team is composed of skilled IT professionals specializing in web-based development, system integration, and process automation. The company focuses on creating customized, scalable, and user-friendly solutions that address real challenges faced by schools and universities — particularly in administrative and academic operations.

Organization Structure



Anonymous
CHIEF EXECUTIVE OFFICER (CEO)



Patricia Mae Puaso
PROJECT MANAGER



Christian Franz Suguitan
BUSINESS ANALYST



Justin Kydel Cervantes
SYSTEM ARCHITECT



Josh Ezekiel Delos Santos
FRONTEND DEVELOPER



Maurik Angelo Fernandez
QUALITY ASSURANCE

Current Business and IT Challenges

Delays in Verification & Approval

Slow processing of student requirements and Memorandums of Agreement (MOAs)

Difficulty in Tracking Student Progress

Inefficient monitoring of document submissions and completion status

Lack of a Centralized Repository

Absence of an accessible system for storing and managing OJT documents

Limited Visibility for Coordinators & Directors

Insufficient monitoring tools to oversee student performance and document flow

Absence of Automated Communication

No system for real-time updates or FAQ assistance for students

At present, Pangasinan State University (PSU) faces several challenges in managing its On-the-Job Training (OJT) operations due to the manual and paper-based process. These issues highlight the urgent need for a streamlined, technology-driven solution that simplifies coordination and improves workflow efficiency within the OJT process.

Purpose of the Proposal

This proposal introduces an Enterprise Architecture (EA) Plan for the development of the PSU OJT Management and Monitoring Portal, an integrated, cloud-based platform designed to modernize the university's OJT processes.

The system aims to establish a seamless digital environment where students, coordinators, and directors can efficiently perform their respective roles — from application and document submission to verification and monitoring. Through the EA approach, the project ensures that the system's design, technology, and operations align with PSU's institutional objectives for digital transformation, transparency, and service quality.

Significance of the Project

The implementation of the PSU OJT Management and Monitoring Portal is a crucial step toward PSU's transition to a fully digital academic management ecosystem. It offers the following benefits:

- **CENTRALIZED REPOSITORY**

  Provides a secure, organized, and easily accessible storage for all OJT-related documents.

- **AUTOMATED WORKFLOW**

  Simplifies the process of submission, verification, and approval through a structured digital system.

- **ENHANCED COMMUNICATION**

  Integrates a chatbot feature to assist students with common inquiries and FAQs

- **DATA TRANSPARENCY AND ACCURACY**

  Ensures that records and reports are reliable, updated, and readily available

- **OPERATIONAL EFFICIENCY**

  - *Reduces redundancy, paperwork, and manual errors*

Overall, this project serves as a model for digital process innovation within the university, enhancing productivity, accessibility, and user satisfaction through the strategic application of enterprise architecture.

II. Conceptual Framework

Enterprise Architecture Overview

Enterprise Architecture (EA) provides a clear structure that connects the university's academic goals with its IT systems. For Pangasinan State University (PSU), EA serves as the foundation in developing the OJT Management and Monitoring Portal. It helps improve organizational efficiency by ensuring that all features, from user roles to document repositories, work together in a systematic and organized way.

EA promotes integration by linking the university's academic and administrative processes into a single platform, while also supporting adaptability by allowing the system to adjust to future technological changes and institutional needs. Through EA, PSU can ensure consistency in its OJT processes, improve communication among stakeholders, and maintain alignment with its goals for digital transformation and service quality.

The Zachman Framework

The Zachman Framework serves as a structured guide for organizing and documenting enterprise components through its well-known 6x6 matrix, which classifies system elements based on six questions (What, How, Where, Who, When, Why) across six stakeholder perspectives (Planner, Owner, Designer, Builder, Subcontractor, and Functioning System).

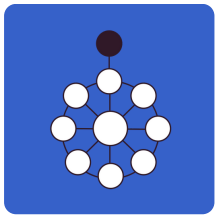
For the PSU OJT Management and Monitoring Portal, the Zachman Framework helps identify and classify the following elements:

Perspective	Description
What	Students, coordinators, directors, Memorandum of Agreement (MOA), evaluation forms, announcements, chatbot, and student repositories
How	The process of student application, submission, verification, and monitoring is managed through automated workflows.
Where	A web-based portal connected to a cloud database for document storage and system access.
Who	Students, coordinators, and directors who perform specific roles such as submitting, verifying, and approving OJT documents.
When	The OJT process timeline, including stages of submission, verification, revision, and approval.
Why	To ensure efficiency, transparency, and accuracy in managing OJT operations while reducing delays and paperwork.

The TOGAF Framework

The TOGAF (The Open Group Architecture Framework) provides the process for developing, governing, and maintaining the architecture of the OJT Management and Monitoring Portal. It focuses on how the system will be designed, built, and improved over time through the Architecture Development Method (ADM)

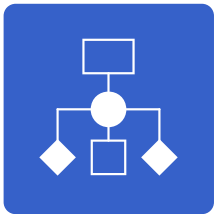
Key components applied in this project include:



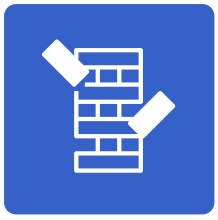
ADM Cycle #1

A structured process that guides the step-by-step design and continuous improvement of the portal to meet PSU's requirements.

Content Framework: #2



Defines the models, diagrams, and deliverables such as use case diagrams, data flow, and process maps that describe system functions for each user role.



Enterprise Continuum: #3

Serves as a collection of reusable components and architecture patterns that can be applied to other university systems.

Governance: #4



Ensures that the system complies with PSU's IT standards, policies, and security measures while maintaining consistency and quality.

Integration of Zachman and TOGAF

Zachman defines what components should be documented, while TOGAF defines how those components should be built and managed. When integrated, Zachman provides the structure and classification, and TOGAF provides the process and methodology.

In the context of PSU's OJT Management and Monitoring Portal, the Zachman Framework can serve as a repository for TOGAF outputs, organizing all deliverables such as process models, data structures, and governance documents. This integration ensures comprehensive coverage from business vision to system implementation.

The result is a well-structured, traceable, and adaptable architecture that supports PSU's goal of digital modernization. Together, Zachman and TOGAF create a balanced approach that strengthens system efficiency, transparency, and sustainability, aligning with the university's mission of continuous improvement and innovation.

III. Objectives of the Project

General Objective

To design and implement a unified Enterprise Architecture for the PSU OJT Management and Monitoring Portal to improve document management, progress tracking, and coordination between stakeholders.

Specific Objectives

1. *To automate the OJT application and verification process.*
2. *To establish a secure, centralized student document repository.*
3. *To enable coordinators and directors to efficiently monitor OJT status.*
4. *To develop a chatbot for student FAQs and assistance.*
5. *To apply the Zachman Framework for structured documentation and the TOGAF ADM for process execution and governance.*

IV. Methodology

A. Data Gathering & Analysis

The data gathering and analysis phase aims to collect relevant information from stakeholders, existing documentation, and organizational systems to form the foundation for developing the enterprise architecture. Multiple techniques will be used to ensure comprehensive and accurate data collection.

Stakeholder Interviews, Workshops, and Surveys:

a. Stakeholder Interview

Interviews will be conducted with key personnel, including system administrators, HR officers, IT staff, faculty coordinators, and OJT supervisors, to identify operational issues, goals, and system requirements.

SAMPLE INTERVIEW GUIDE

Section	Sample Questions
Introduction	• Briefly explain your role in the OJT process. • How does your department currently monitor OJT students?
Current Process	• What are the main challenges in tracking OJT activities? • How do you currently manage student records and company partnerships?
Information Systems	• What tools or software do you use in your daily OJT-related tasks? • Are there any limitations or problems with these systems?
Future Expectations	• What features or improvements would you like to see in a new OJT Management System? • How can technology improve coordination between students, companies, and coordinators?

SAMPLE INTERVIEW GUIDE

- *Semi-structured interview format to allow flexibility.*
- *Each interview will last approximately 30–45 minutes.*
- *Data will be recorded via notes and audio (with consent).*
- *Interviews will be scheduled through coordination with department heads.*

b. Stakeholder Workshops

Workshops will be conducted to validate gathered data and refine process diagrams (e.g., BPMN, DFD). Participants will include representatives from IT Services, HR, and OJT offices. These workshops will help establish:

- *Consensus on current ("As-Is") processes.*
- *Identification of pain points and improvement opportunities.*
- *Validation of proposed ("To-Be") architecture views.*

c. Survey Distribution

A structured online survey will be administered to gather quantitative feedback from a larger audience (students, faculty, and OJT coordinators).

SAMPLE INTERVIEW GUIDE

Question	Response Type
1. How often do you experience difficulty submitting or tracking OJT requirements?	□ Never □ Rarely □ Sometimes □ Often □ Always
2. How satisfied are you with the current OJT monitoring system/process?	□ Very Satisfied □ Satisfied □ Neutral □ Dissatisfied □ Very Dissatisfied
3. Which system feature would be most helpful?	□ Online submission □ Real-time tracking □ Coordinator feedback □ Company evaluation portal
4. What improvements would you suggest for the OJT process?	*Open-ended response*

SAMPLE INTERVIEW GUIDE

- *Hosted via Google Forms or Microsoft Forms for accessibility.*
- *Distributed through email and student group chats.*
- *Target respondents: ~100 participants (students, coordinators, and supervisors).*
- *Data will be summarized statistically using frequency and percentage analysis.*

Document Collection and Review

To complement stakeholder inputs, existing documentation and resources will be gathered to provide context and validation for process modeling and architecture design.

DOCUMENTS TO BE COLLECTED

- *Organizational chart of the IT department and OJT offices.*
- *Existing process manuals and standard operating procedures.*
- *Previous OJT tracking forms and templates.*
- *Current system inventory and software documentation (if any).*
- *Data privacy and security policy documents.*

COLLECTION STRATEGY

- *Coordinate with department heads and document custodians for access.*
- *Use document request forms and official email correspondence to ensure authorization.*
- *Review documents systematically using a Document Review Matrix, identifying relevance, completeness, and consistency with observed processes.*
- *Extract process steps, actors, and data entities to support the development of BPMN, DFD, and ERD diagrams.*

Document Collection and Review

- ***Qualitative Analysis:*** *Thematic coding of interview transcripts to identify common issues and goals.*
- ***Quantitative Analysis:*** *Aggregation of survey responses to measure stakeholder satisfaction and identify priority improvements.*
- ***Synthesis:*** *Integration of findings into the Architecture Vision and Business Architecture phases of the TOGAF ADM.*

B. Application of Zachman Framework

Information will be classified according to Zachman's six interrogatives (What, How, Where, Who, When, Why) across stakeholder perspectives (Planner, Owner, Designer, Builder).

Perspective	What	How	Where	Who	When	Why
Planner	Student data	Application flow	Web portal	Users	Submission timeline	Improve efficiency
Owner	MOA Documents	Verification	Cloud storage	Coordinators , Director	Approval phase	Ensure compliance
Designer	Database Schema	System modules	Hosting server	Developers	Development phase	Integration
Builder	Data fields	APIs, UI design	Server-side	Dev Team	Testing schedule	Implementation
User	Records	Upload acions	Dashboard	Students	Deadlines	Completion & tracking
Maintenanc e / Change Manager	Updated records	Workflow revisions	EA Repository	Admin / QA Officer	Continuous update cycle	Governance improvement

C. Application of TOGAF ADM Phases with Techniques & Diagrams

ADM Phase	Activities	Diagrams/Techniques
Preliminary Phase	Define EA principles, assign governance roles	EA Governance Diagram, Stakeholder Map
Architecture Vision	Define project goals, scope, and stakeholders	Value Chain Diagram, Capability Map
Business Architecture	Model business processes and OJT workflow	BPMN Diagram, Use Case Diagram
Information Systems Architecture	Model data and system relationships	DFD, ERD
Technology Architecture	Design hosting, database, and security setup	Network Topology Diagram, Deployment Diagram
Opportunities & Solutions	Identify enhancement areas	Gap Analysis Matrix, Solution Architecture Diagram
Migration Planning	Plan phase-wise implementation	Roadmap Diagram, WBS
Implementation Governance	Monitor compliance and execution	Compliance Assessment Matrix, Governance Workflow
Architecture Change Management	Establish EA update procedures	Architecture Maintenance Process Diagram

C. Application of TOGAF ADM Phases with Techniques & Diagrams

PRELIMINARY PHASE

ARCHITECTURE PRINCIPLES

- **Simplicity First :** Build modular, maintainable architecture suitable for student-managed deployment.
- **Security by Design** : Use Firebase Auth and cloud storage access rules.
- **Centralized Repository** : All student documents are stored securely in one structured drive-like repository.
- **Scalability for Future Use:** Can be extended to other PSU campuses or departments.
- **Transparency and Traceability:** Every document upload and approval is logged.
- **User-Centric Design:** Focus on clean UI for students and coordinators.
- **Collaborative Development:** Each member contributes a major subsystem or documentation phase.
- **Continuous Improvement:** Regular internal reviews to improve features and EA alignment.

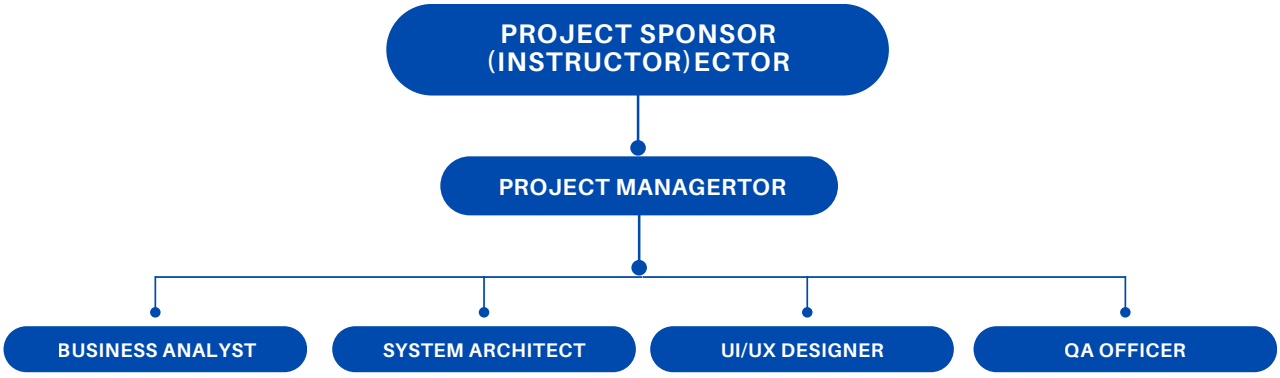
PRELIMINARY PHASE

GOVERNANCE ROLES

Role	Responsibilities
Project Manager / EA Lead	Oversees project, ensures alignment with EA principles, manages timeline and deliverables.
Business Analyst / Documentation Lead	Gathers requirements, conducts interviews/surveys, documents current and target architectures.
System Architect / Developer	Designs system modules, database schema, and ensures compliance with EA frameworks (Zachman + TOGAF).
UI/UX Designer / Frontend Developer	Designs and develops user interfaces for Students, Coordinators, and Directors.
Quality Assurance / Security & Compliance Officer	Tests system functions, ensures data protection, conducts user acceptance testing and feedback collection.

PRELIMINARY PHASE

ARCHITECTURE GOVERNANCE STRUCTURE



STAKEHOLDER MAP

Stakeholder	Category	Interest / Concern	Involvement Level
Students	External	Ease of use, document upload, clear status	High
OJT Coordinators	External	Workflow efficiency, tracking, communication	High
Director	External	Review, approve, and sign documents efficiently	Medium
University IT Office	External	Integration, data security	Medium
Project Team	Internal	System design, documentation, governance compliance	High
Instructor / Adviser	Internal	Oversight, grading, and project approval	High

ARCHITECTURE VISION

HIGH-LEVEL GOALS

Goal No.	Goal Statement	Description / Expected Outcome
G1	**Digitize OJT management process**	*Transform the manual, paper-based OJT management process into a centralized digital platform accessible by students, coordinators, and the director.*
G2	**Enhance transparency and monitoring**	*Allow all stakeholders to track MOA, evaluation, and document status in real-time.*
G3	**Streamline document verification and approval**	*Implement a structured approval workflow for coordinators and directors, reducing delays and redundancy.*
G4	**Establish centralized document repository**	*Maintain an organized, secure repository where each student's OJT documents are stored, retrieved, and updated easily.*
G5	**Enable communication and announcements**	*Provide coordinators with tools to post memos, updates, and reminders to students within the system.*
G6	**Integrate chatbot for FAQs and guidance**	*Support students with an intelligent chatbot that can handle frequently asked queries regarding OJT procedures and requirements.*
G7	**Support governance and scalability**	*Define governance principles and architecture standards to ensure sustainability and future system expansion.*

ARCHITECTURE VISION

SCOPE OF THE ENTERPRISE ARCHITECTURE PROJECT

Aspect	Scope Definition
Business Scope	*Covers the end-to-end OJT lifecycle — from student registration and document submission to coordinator verification and director approval. Includes communication (announcements, chatbot) and system administration.*
Functional Scope	*Implementation of user authentication, document upload modules, MOA verification workflows, announcements module, chatbot integration, and governance features (role management, compliance).*
Organizational Scope	*Applies to PSU stakeholders: Students, OJT Coordinators, and Director, with the development and maintenance managed by the project team (5 members).*
Technology Scope	*Web-based portal hosted on a cloud server, with integrated document repository, database schema for user and submission data, and chatbot powered by rule-based or AI-driven logic.*
Geographic Scope	*Initially deployed for the Pangasinan State University main campus, scalable to other campuses in future phases.*
Out of Scope	*Offline processing, external company-side evaluations beyond document uploads, or integration with third-party LMS.*

ARCHITECTURE VISION

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ARCHITECTURE VISION

BUSINESS CAPABILITY MAP

Business Capability Map – PSU OJT Management and Monitoring Portal

LEVEL 1: CORE CAPABILITIES

STUDENT MANAGEMENT

01

- *Account Registration*
- *Student Profiling*
- *Document Upload (MOA, Eval)*
- *Progress Tracking*

OJT PROCESS MANAGEMENT

02

- *MOA Verification Workflow*
- *Status Update (Processing, Revise, For Signature)*
- *Approval and Sign-off*
- *Repository*

COMMUNICATION & SUPPORT

03

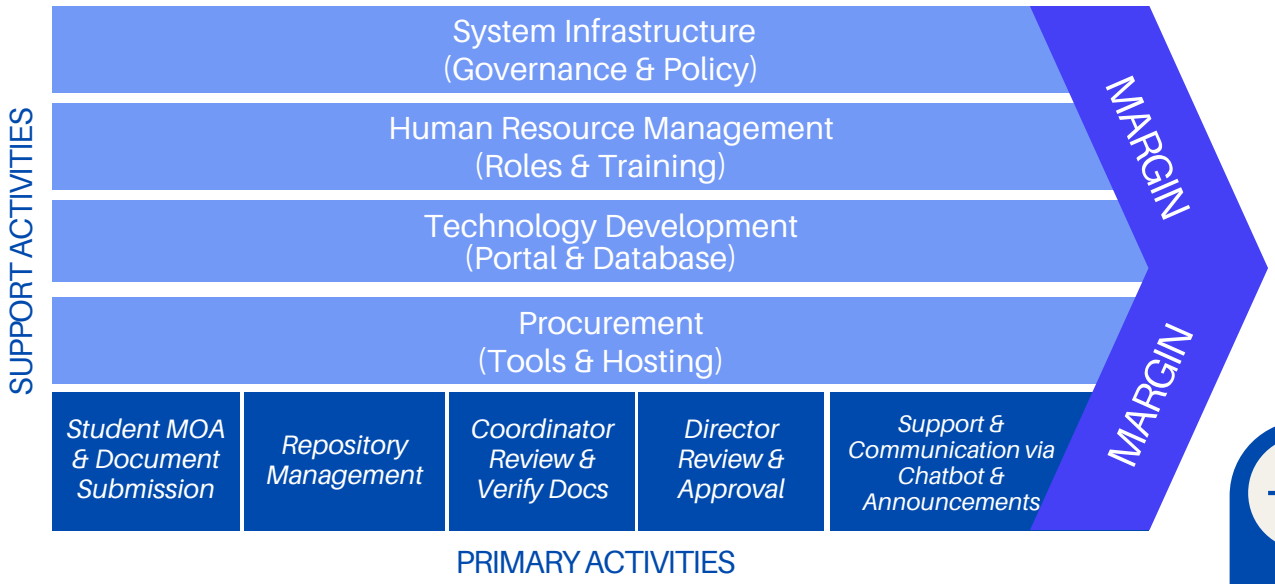
- *Announcements & Memos*
- *Chatbot Assistant (FAQ)*
- *Coordinator Comments*
- *Notification Alerts*

SYSTEM ADMIN

04

- *Role and Permission Mgmt*
- *Coordinator Account Setup*
- *Data Monitoring & QA*
- *EA Governance Oversight*

VALUE CHAIN DIAGRAM



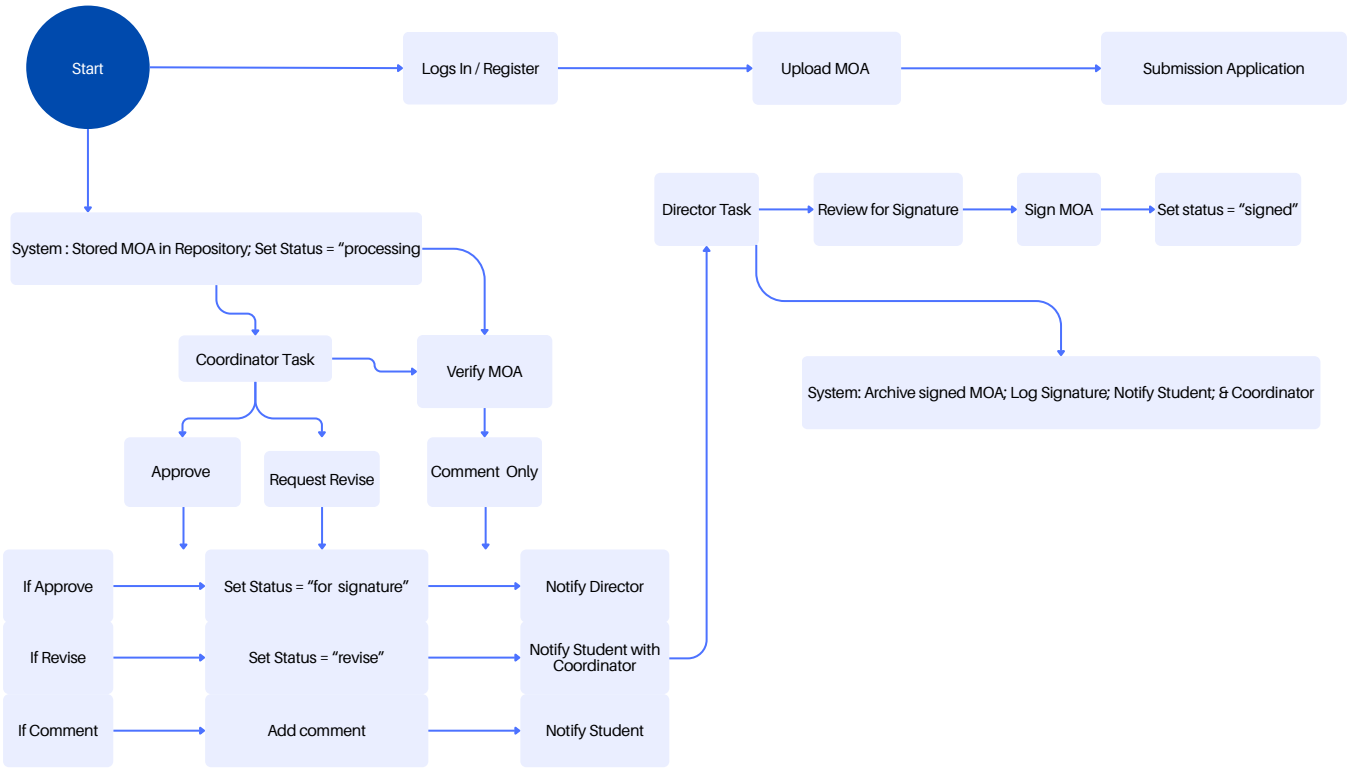
BUSINESS ARCHITECTURE

KEY BUSINESS FUNCTIONS

Function	Description
Student Registration	*Students create accounts, upload required OJT documents, and update their profiles.*
Coordinator Review	*Coordinators verify student submissions, provide comments, and request revisions when needed.*
Director Approval	*Directors confirm and digitally sign approved OJT-related documents.*
Repository Access	*Centralized storage for all validated documents, ensuring secure and ongoing access.*
Announcement Management	*Coordinators post announcements, reminders, events, and important updates for students.*
Chatbot Support	*An AI assistant provides automated help with FAQs, deadlines, and navigation assistance within the portal.*
Role Management	*Administrators define and manage system roles and permissions (e.g., Student, Coordinator, Director, QA).*

BUSINESS ARCHITECTURE

BUSINESS PROCESS MODEL AND NOTATION (BPMN)



INFORMATION SYSTEMS ARCHITECTURE

DATA ARCHITECTURE

The Data Architecture identifies the core data entities and their relationships to support OJT operations. Key data entities include Students, Coordinators, Directors, MOA Documents, Evaluation Reports, Announcements, and System Logs. These entities are organized within a centralized cloud database to eliminate data redundancy and ensure consistency across the system.

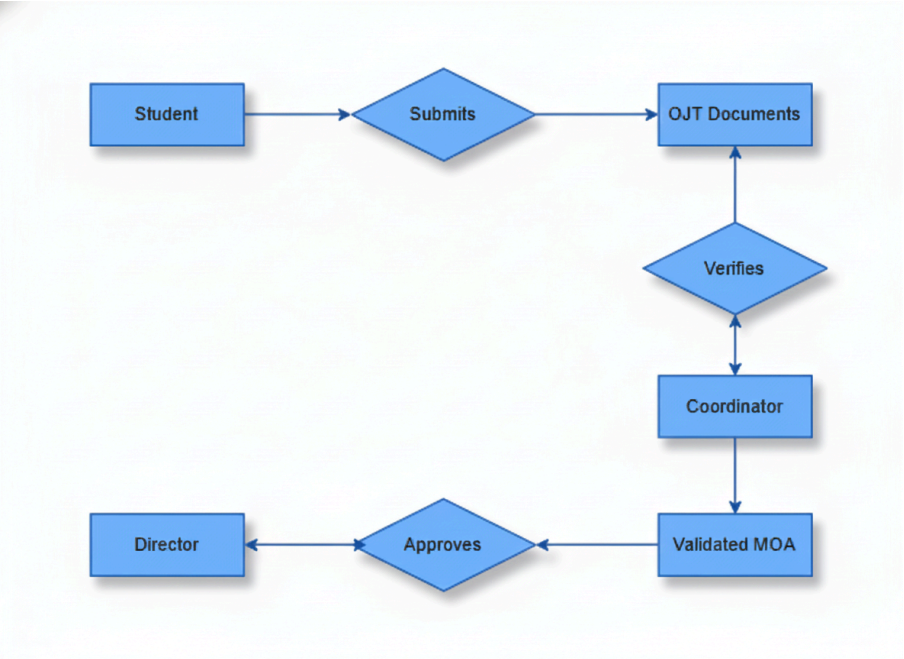
APPLICATION ARCHITECTURE

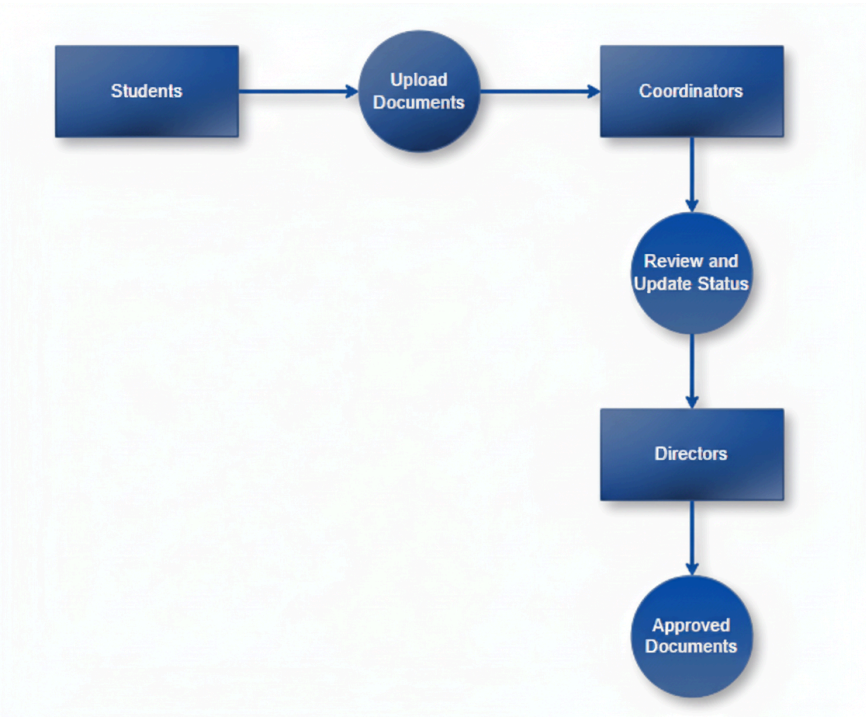
The Application Architecture defines the logical structure of system modules and their interactions. The PSU OJT portal is composed of the following core modules:

- User Authentication and Role Management
- Student Document Submission and Tracking
- MOA Verification and Approval Workflow
- Announcement and Notification Module
- Chatbot Support Module
- Centralized Document Repository

INFORMATION SYSTEMS ARCHITECTURE

ENTITY RELATIONSHIP DIAGRAM AND DATA FLOW DIAGRAM





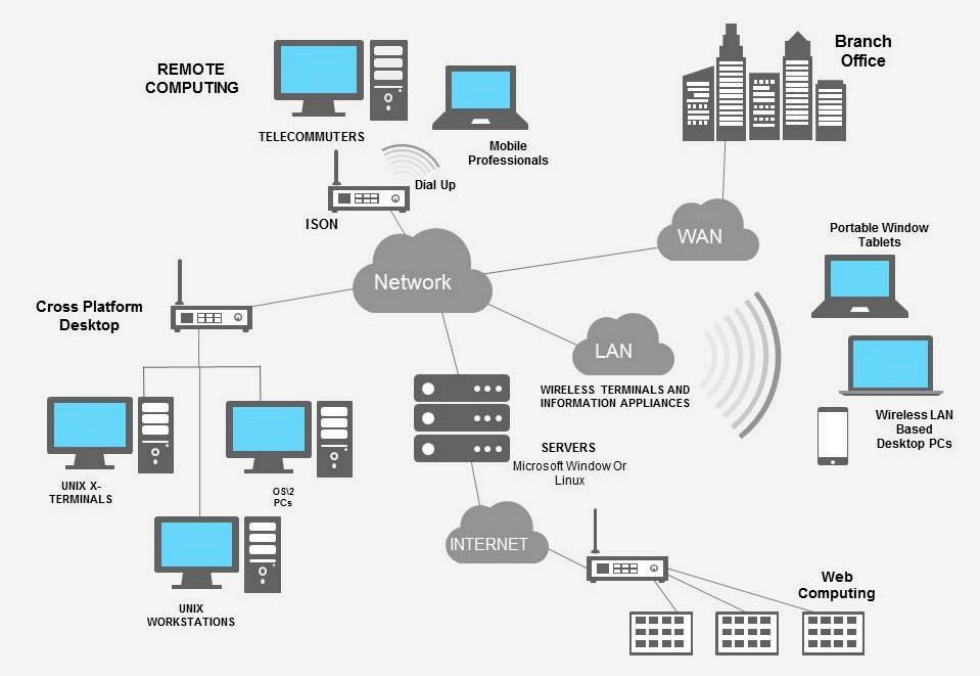
TECHNOLOGY ARCHITECTURE

The Technology Architecture phase defines the technical infrastructure required to deploy and operate the PSU OJT Management and Monitoring Portal securely and efficiently.

The system is designed as a web-based, cloud-hosted application accessible through standard web browsers. The technology stack includes:

- Cloud hosting services for application deployment
- Cloud database and storage services for centralized document management
- Secure authentication mechanisms using role-based access control
- API-driven backend to support scalability and system integration

NETWORK TOPOLOGY DIAGRAM



OPPORTUNITIES AND SOLUTIONS

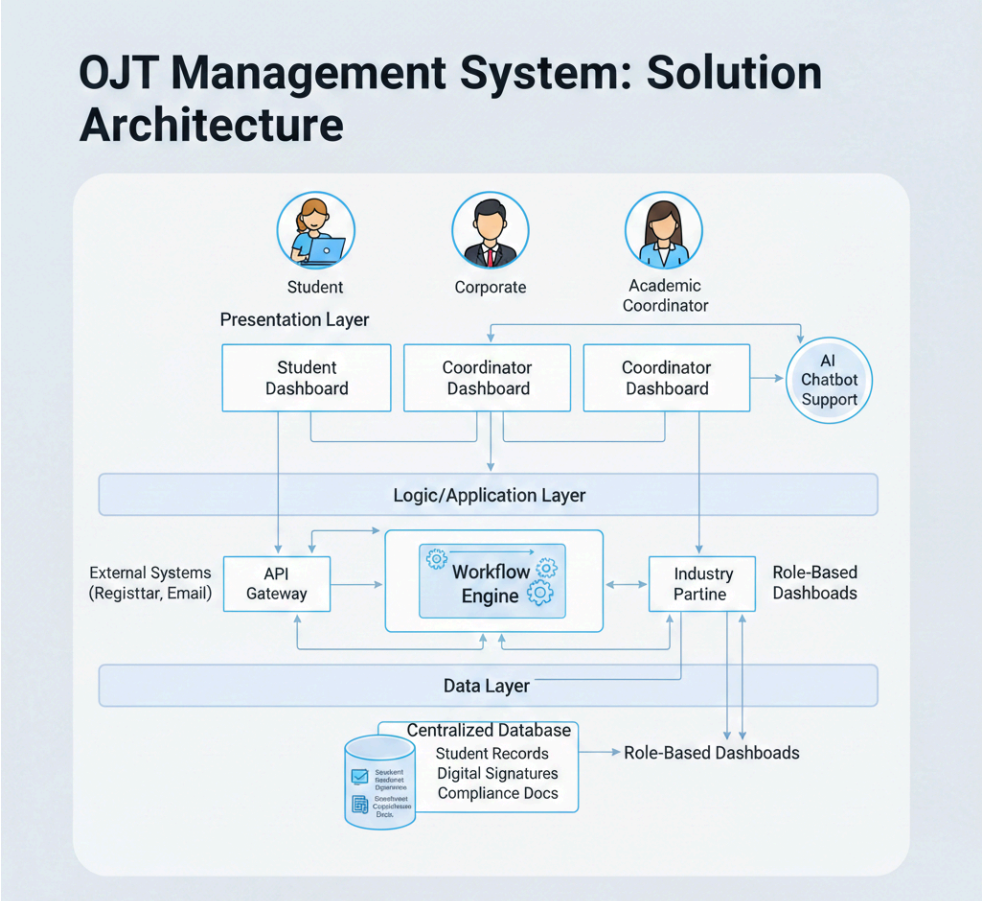
The Opportunities and Solutions phase identifies improvement areas based on gaps between the current ("As-Is") and target ("To-Be") architectures.

GAP ANALYSIS MATRIX

Focus Area	Current State	Future State	**Identified gap**	**Required Action Solution**
Submission Process	*Manual: Paper-based or email attachments; prone to delays and human error.*	*Automated: Seamless online portal with validation checks.*	*Lack of a digital intake interface.*	*Implement a web-based Online Submission Portal.*
Data Storage	*Disparate: Files scattered across local drives, physical folders, and email inboxes.*	*Centralized: Single source of truth via a secure cloud environment.*	*Fragmented data architecture and high retrieval time.*	*Migrate all assets to a Centralized Cloud Repository (e.g., SharePoint, AWS, or Google Drive).*
Oversight	*Limited Visibility: Hard to track status; requires manual follow-ups to find bottlenecks.*	*Real-Time Visibility: Instant insights into the lifecycle of every document.*	*Absence of live data visualization.*	*Develop a Real-Time Tracking Dashboard with KPI reporting.*
Communication	*None/Manual: Support is reactive; no proactive updates for users.*	*Integrated & Proactive: Instant support and automated status triggers.*	*Manual touchpoints for routine inquiries.*	*Deploy an Integrated Chatbot and an Automated Notification Engine.*

OPPORTUNITIES AND SOLUTIONS

SOLUTION ARCHITECTURE DIAGRAM



MIGRATION PLANNING

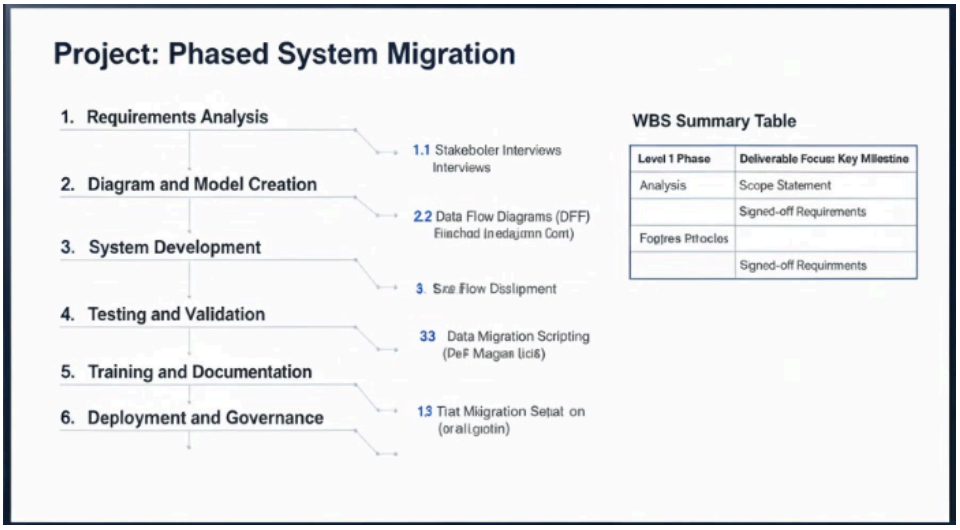
The Migration Planning phase defines the transition strategy from the current manual OJT process to the fully digital PSU OJT Management and Monitoring Portal.

ROADMAP DIAGRAM



MIGRATION PLANNING

WORK BREAKDOWN STRUCTURE



ADM PHASES

IMPLEMENTATION GOVERNANCE

The Implementation Governance phase ensures that the development and deployment of the PSU OJT Management and Monitoring Portal comply with the defined enterprise architecture, governance principles, and institutional policies. This phase focuses on monitoring project execution, validating architectural compliance, and ensuring that system implementation aligns with approved designs and standards.

COMPLIANCE ASSESSMENT MATRIX

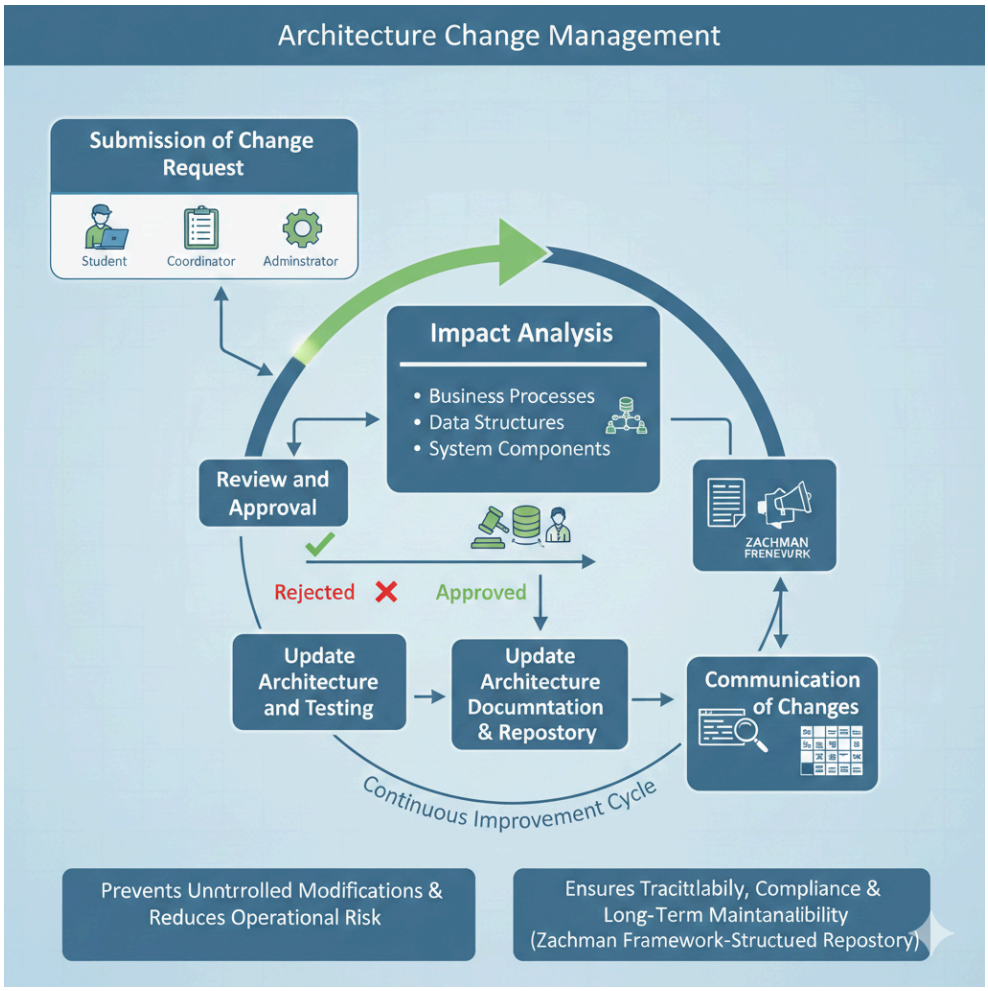
Governance Domain	Compliance Requirement	Compliance Status	**Evidence/Observation**	**Corrective Action (if non-compliant)**
Architectural Alignment	*Adherence to the defined 3-Tier Architecture (Client, Server, Database).*	*[Compliant / Non / Partial]*	*Review of system topology and API structure.*	*Re-factor services to decouple logic from the UI layer.*
Data Privacy (RA 10173)	*Encryption of student personal data (PII) at rest and in transit.*	*[Compliant / Non / Partial]*	*SSL/TLS implementation check and database encryption status.*	*Implement AES-256 encryption for the student database.*
User Access Control	*Role-Based Access Control (RBAC) implementation (OJT, Coordinator, Host).*	*[Compliant / Non / Partial]*	*Audit of authentication logs and permission tables.*	*Update the Auth module to restrict Coordinator views.*
Data Integrity	*Real-time logging of OJT hours and supervisor validations.*	*[Compliant / Non / Partial]*	*Verification of audit trail logs and timestamping.*	*Integrate automated server-side time validation.*

ADM PHASES

ARCHITECTURE CHANGE MANAGEMENT

The Architecture Change Management phase establishes formal mechanisms for managing system changes, updates, and continuous improvement after deployment. This phase ensures that the enterprise architecture remains relevant, controlled, and aligned with PSU's evolving operational and technological needs.

ARCHITECTURE MAINTENANCE PROCESS DIAGRAM



USERS INFOGRAPHICS



D. Integration Plan

Zachman Perspective	TOGAF ADM Phase Source	Exact TOGAF Diagrams / Techniques to Store	Purpose in Repository
Planner	*Preliminary Phase, Architecture Vision*	*- EA Governance Diagram - Stakeholder Map - Value Chain Diagram - Business Capability Map - Architecture Vision Document*	*Establishes the architecture's foundation, guiding principles, and high-level goals.*
Owner	*Business Architecture*	*- Business Process Model and Notation (BPMN) - Use Case Diagrams - Organizational Structure Diagram*	*Documents how the business operates and how OJT workflows support goals.*
Designer	*Information Systems Architecture*	*- Data Flow Diagram (DFD) - Entity-Relationship Diagram (ERD)*	*Defines logical data design and relationships between system components.*
Builder	*Technology Architecture*	*- Network Topology Diagram - Deployment Diagram*	*Represents the physical and technical infrastructure of the system.*
User	*Opportunities & Solutions, Migration Planning*	*- Solution Architecture Diagram - Gap Analysis Matrix - Roadmap Diagram - Work Breakdown Structure (WBS)*	*Shows system rollout, implementation plan, and improvement roadmap.*
Maintenance / Change Manager	*Implementation Governance, Architecture Change Management*	*- Compliance Assessment Matrix - Governance Workflow - Architecture Maintenance Process Diagram*	*Ensures EA compliance, continuous improvement, and version tracking.*

V. Expected Deliverables

1) ENTERPRISE ARCHITECTURE BLUEPRINT (CURRENT AND TARGET STATES)

A visual and descriptive model representing the present OJT workflow and the proposed digital architecture for improved process automation and monitoring.

2) ZACHMAN-BASED ARCHITECTURE REPOSITORY

A centralized documentation library structured according to the Zachman Framework, organizing artifacts by the six interrogatives (What, How, Where, Who, When, Why) and stakeholder perspectives.

3) TOGAF ADM PROCESS DOCUMENTATION AND ROADMAP

Complete documentation of the architecture development process, including diagrams, models, and implementation plans following TOGAF's ADM phases.

4) GOVERNANCE AND CHANGE MANAGEMENT PLAN

A framework outlining governance roles, approval workflows, and procedures for maintaining and updating the system architecture over time.

5) USER TRAINING MANUAL AND ADOPTION GUIDE

Training materials and manuals designed to help students, coordinators, and directors transition smoothly to the new portal and utilize its features effectively.

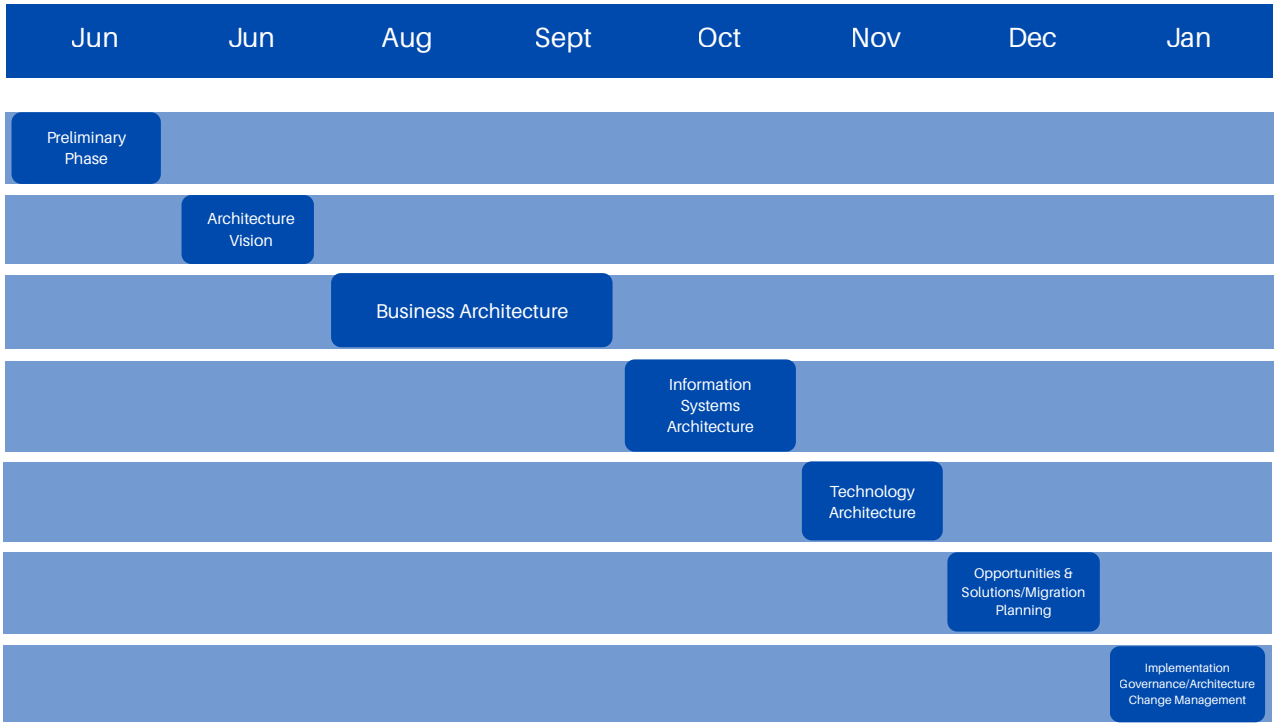
Deliverable 1:Enterprise Architecture Blueprint (Current and Target States)

CURRENT STATE ***(As-Is Architecture)***	***TARGET STATE*** ***(To-Be Architecture)***	
Manual OJT coordination using physical documents and email-based submissions	*Automated OJT coordination through an online portal integrating coordinators, directors, and students*	*Business Architecture*
Student and MOA records are stored in spreadsheets and departmental drives	*Centralized cloud database managing student, MOA, and progress data*	*Data Architecture*
No integrated platform — reliance on shared drives, forms, and manual processing	*Web-based OJT Management System with role-based access and document tracking features*	*Application Architecture*
Uses basic web storage and local networks; minimal automation tools	*Secure cloud deployment using web hosting and API-driven backend infrastructure*	*Technology Architecture*

Deliverable 2: Repository of architecture diagrams and documentation (Zachman-structured)

Perspective	What (Data)	How (Function)	Where (Network)	Who (People)	When (Time)	Why (Motivation)
Planner	*List of OJT data entities (students, MOAs)*	*Overview of OJT process*	*Cloud-based web portal*	*Students, coordinators directors*	*OJT timeline and deadlines*	*Improve efficiency and transparency*
Owner	*Document templates, student profiles*	*Workflow diagrams (submission, verification approval)*	*PSU campus network*	*OJT coordinators and directors*	*Phase schedules*	*Ensure compliance and accuracy*
Designer	*Database schema, entity relationships*	*System modules, process logic*	*Hosting and storage architecture*	*Developers, system analysts*	*Development milestones*	*System integration and optimization*
Builder	*Data fields, API mappings*	*Backend and frontend code*	*Server infrastructure*	*Development team*	*Testing schedule*	*Implement automated system*
Sub-Contractor	*SQL scripts, JSON configs*	*Program flow, UI/UX layout*	*Deployment pipeline*	*QA engineers*	*Release plan*	*Maintain quality assurance*
Functioning System	*Active data records*	*Executing workflows*	*Live portal environment*	*End users (students, staff)*	*Real-time updates*	*Continuous improvement*

Deliverable 3: TOGAF ADM-based Architecture Roadmap



Deliverable 4: Governance and Change Management Plan.

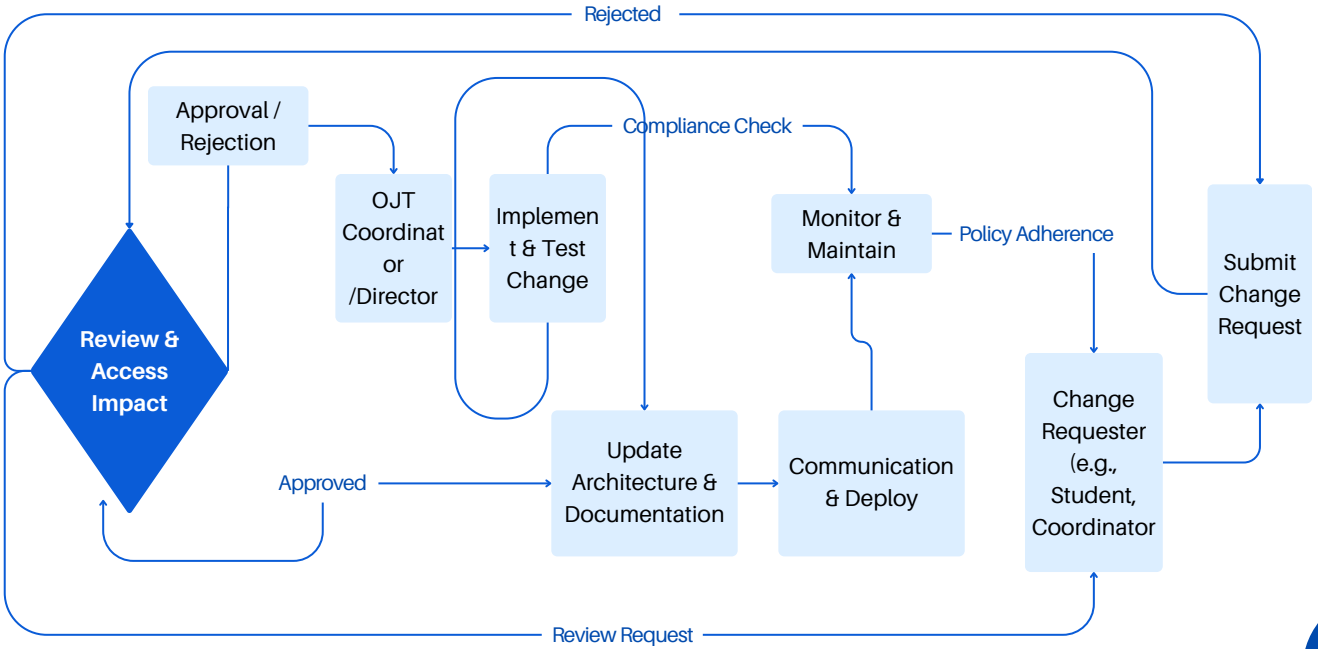
Governance and Change Management Plan

To ensure the stability, transparency, and long-term sustainability of the PSU OJT Management and Monitoring Portal, a structured Governance and Change Management Plan is implemented. This plan establishes a systematic approach for reviewing, approving, and maintaining architectural changes in alignment with Pangasinan State University's IT governance principles.

Activity:

Model the governance and change control processes of the OJT Management Portal, including the roles of coordinators, directors, and administrators in evaluating, approving, and maintaining system updates and documentation. This activity ensures that all modifications undergo formal review and compliance validation before implementation.

Governance Workflow Diagram



Deliverable 5: Stakeholder Training and EA Adoption Guide.

Stakeholder Training and EA Adoption Guide

The success of the PSU OJT Management and Monitoring Portal depends on effective stakeholder engagement and adoption. A structured training and adoption guide ensures that all users — students, coordinators, and directors — can efficiently use the portal and comply with the established governance processes.

Activity:

Model the stakeholder engagement and training process, defining user roles, adoption phases, and support mechanisms to ensure effective system utilization and compliance. This includes pilot testing, continuous feedback integration, and capacity-building programs for long-term sustainability.

Deliverable 5: Stakeholder Training and EA Adoption Guide.

Stakeholder Training and EA Adoption Guide

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Stakeholder Engagement Matrix

Stakeholder Group	Role in OJT Portal	Engagement Level (I/C/A)	Key Training Focus
Students (OJT Trainees)	*Submit documents, log hours, and track status.*	*I - Informed, C - Consulted, A - Approved*	*Portal navigation, document upload process, and status tracking.*
OJT Coordinators (Department)	*Approve documents, monitor student progress, and evaluate submissions.*	*C, A*	*Workflow approval process, compliance verification, and user support.*
Directors/Deans (College)	*Provide oversight and final approval of OJT plans and reports.*	*A*	*Dashboard review, report validation, and decision-making.*
EA Administrator (OJT Office)	*Manage architecture repository, updates, and system compliance.*	*C, A*	*System governance, change management, and compliance tracking.*
IT Support Staff (IT Office / Helpdesk)	*Provide technical assistance and resolve user issues.*	*I, C*	*System troubleshooting, security maintenance, and user FAQs.*

VI. Benefits & Impact

The implementation of the PSU OJT Management and Monitoring Portal will deliver significant benefits to students, coordinators, and directors by transforming manual OJT procedures into an automated and transparent digital process. The system's enterprise architecture ensures efficiency, accessibility, and sustainability across all operational levels.

SIMPLIFIED AND FASTER OJT DOCUMENT SUBMISSION

The portal enables students to submit all required documents — such as MOAs, evaluations, and reports — through a user-friendly online platform. This automation reduces the time spent on manual submissions, minimizes physical paperwork, and ensures faster processing of requirements.

TRANSPARENCY IN VERIFICATION AND MONITORING PROCESSES

Coordinators and directors can track each document's progress in real time, with status indicators such as processing, for revision, or for signature. This transparency fosters accountability and allows students to stay informed about their application status at every stage.

REDUCED WORKLOAD FOR COORDINATORS AND DIRECTORS

Automated workflows simplify verification, document validation, and progress tracking, lessening the administrative burden on staff. This allows coordinators and directors to focus more on supervision, evaluation, and guidance rather than repetitive clerical tasks.

INSTANT ACCESS TO DATA VIA CENTRALIZED REPOSITORY

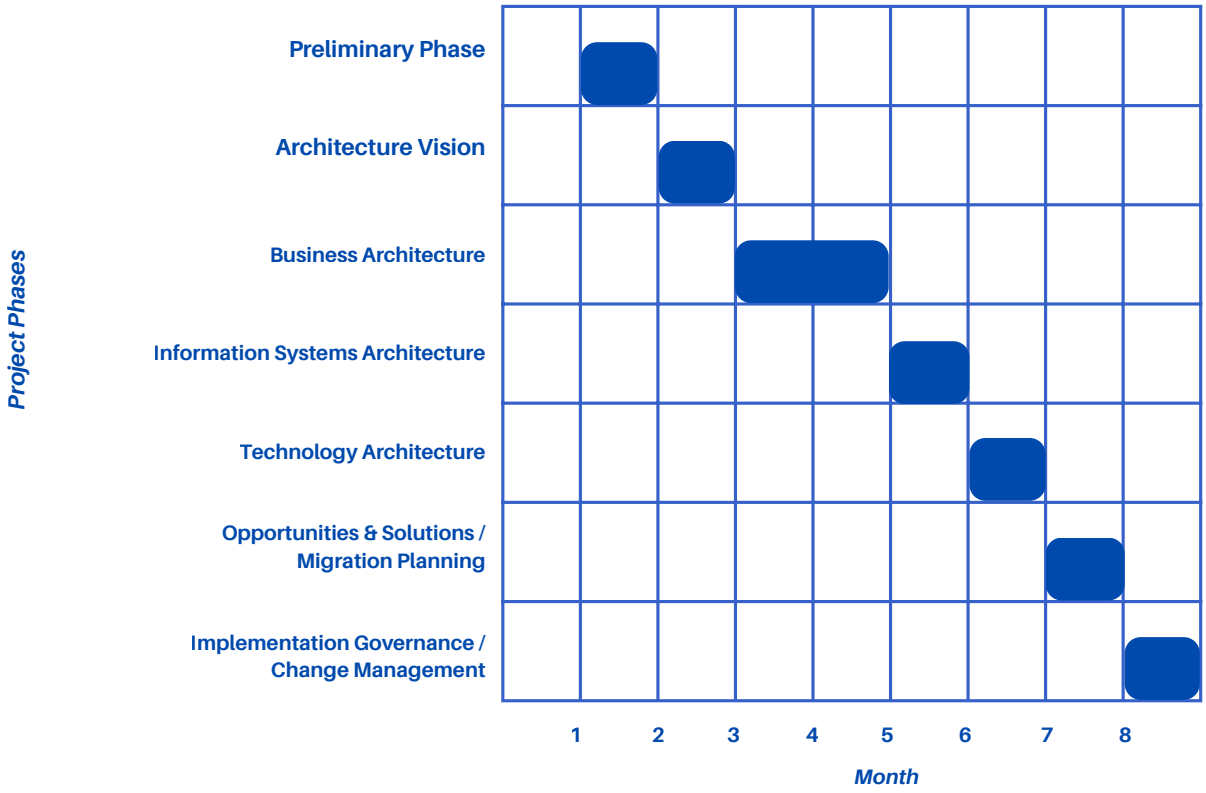
All OJT-related files are securely stored in a centralized, cloud-based repository, making them easily retrievable when needed. This organized data management promotes consistency, data integrity, and accessibility for both students and university personnel.

IMPROVED STUDENT SATISFACTION AND INSTITUTIONAL EFFICIENCY

The integration of a chatbot and digital workflows enhances communication, responsiveness, and overall service quality. Students experience faster feedback and a smoother process, while the university benefits from a more efficient, transparent, and data-driven OJT management system.

VII. Project Timeline

GANTT CHART OF THE SEQUENCE OF ADM PHASES & DIAGRAM CREATION SCHEDULE



VIII. Budget Estimate

The estimated budget for the development and implementation of the PSU OJT Management and Monitoring Portal reflects the resources needed to design, document, and operationalize the enterprise architecture using the Zachman and TOGAF frameworks. The proposed allocation supports both technical and organizational requirements to ensure a sustainable and well-governed deployment.

Category	Description	Estimated Cost (₱)	Remarks
1. Training Sessions for EA Modeling	*Hands-on workshops and seminars on TOGAF ADM, Zachman Framework, and EA tools (e.g., BPMN, DFD, ERD modeling).*	₱25,000	*Includes materials, software access, and facilitator fees.*
2. Consultant / Analyst Fees	*Hiring or compensating EA consultants or system analysts for technical guidance, architecture validation, and framework integration.*	₱40,000	*Based on professional service rate (₱5,000/month for 8 months).*
3. Documentation and Governance Setup	*Preparation of EA documentation, diagrams, and governance framework setup (repository, compliance templates).*	₱15,000	*Includes printing, cloud storage subscription, and repository configuration.*
4. Software Tools and Hosting	*Cloud hosting for EA repository, version control, and modeling tools (e.g., Lucidchart, Draw.io, Firebase, Render).*	₱10,000	*Covers 8-month subscription and maintenance.*
5. Contingency Fund	*Reserved for unexpected project expenses (e.g., system upgrades, additional tools, internet cost).*	₱5,000	10% of total allocation.
TOTAL ESTIMATED BUDGET		**₱95,000.00**	**Estimated for full 8-month EA implementation**

The total estimated cost of ₱95,000 will cover both the enterprise architecture design and the system development components, ensuring the portal's functionality, sustainability, and institutional readiness.

IX. Risk Management

Effective risk management is essential in ensuring the successful implementation and sustainability of the PSU OJT Management and Monitoring Portal. The following table identifies potential risks, their impact, and the corresponding mitigation strategies aligned with the project's enterprise architecture governance plan.

Risk	Impact Level	Mitigation Strategy
Resistance to the New System	Medium	*Conduct stakeholder orientation and pilot testing to familiarize users (students, coordinators, and directors) with the portal's interface and features. Provide continuous user support through FAQs and chatbot integration.*
Data Inconsistency and Loss	High	*Implement automated data validation and regular backup procedures within the cloud repository. Establish clear data governance policies to maintain accuracy and integrity.*
System Downtime and Accessibility Issues	Medium	*Utilize faculty mentorship and cloud-based hosting services to reduce dependency on local servers. Engage IT experts and student developers for technical support and maintenance*
Contingency	Medium	*Schedule periodic maintenance and establish a failover setup to minimize downtime. Deploy monitoring tools to track system performance and ensure 24/7 accessibility.*
Cybersecurity Threats	High	*Employ encryption, secure authentication, and role-based access controls. Conduct regular vulnerability assessments to protect sensitive student and institutional data.*

By proactively addressing these risks, the project will maintain operational continuity, data reliability, and user confidence, ensuring that the enterprise architecture remains aligned with PSU's digital transformation objectives.

X. Conclusion & Recommendation

Conclusion

The integration of the Zachman Framework and the TOGAF Architecture Development Method (ADM) establishes a strong foundation for the PSU OJT Management and Monitoring Portal. Through Zachman's structured documentation and TOGAF's systematic process, the project ensures a well-governed, traceable, and scalable enterprise architecture that addresses the university's current operational challenges in OJT management.

The proposed portal not only modernizes the documentation and verification workflow but also enhances transparency, accountability, and efficiency through automation, centralized repositories, and intelligent support tools such as chatbots. These features collectively contribute to improved student experience and streamlined coordination between students, OJT coordinators, and directors.

Recommendation

1) INSTITUTIONAL ADOPTION AND POLICY INTEGRATION

Pangasinan State University should formally adopt the PSU OJT Management and Monitoring Portal as part of its academic digitalization strategy. Policies must be updated to support electronic document submissions, digital verification, and repository-based record-keeping.

2) CONTINUOUS SYSTEM IMPROVEMENT

Regular system evaluations should be conducted every academic term to identify improvement areas, such as user interface enhancements, new features, and performance optimization. Feedback mechanisms from students and coordinators should guide these updates.

3) CAPACITY BUILDING AND TRAINING PROGRAMS

Conduct continuous training for coordinators, directors, and administrative staff to ensure efficient system utilization. Workshops and orientations will improve digital literacy and promote confidence in using the portal's features.

4.) DATA SECURITY AND COMPLIANCE MEASURESNT PLAN

Implement a strict data governance framework following the Data Privacy Act of 2012 (RA 10173). Regular security audits and encryption protocols should be enforced to protect sensitive student and institutional data.

5) INTEGRATION WITH OTHER UNIVERSITY SYSTEMS

Future system upgrades may include linking the OJT portal with PSU's existing student information system (SIS), HR, and registrar databases to create a unified digital ecosystem for academic management.

6. SCALABILITY FOR MULTI-CAMPUS USE

The architecture should be scalable to accommodate multiple PSU campuses, ensuring that all students, regardless of location, can access the same digital OJT management experience.

Recommendation

7) DEDICATED SUPPORT AND MAINTENANCE TEAM

Establish a technical support and maintenance team within the IT department to monitor system health, manage updates, and address user concerns promptly.

8) SUSTAINABILITY THROUGH CLOUD AND OPEN-SOURCE SOLUTIONS

Adopt cost-efficient, cloud-based, and open-source technologies to ensure long-term sustainability and lower maintenance costs for the university.

9) PERIODIC REVIEW OF ENTERPRISE ARCH

Conduct an annual review of the EA documentation (Zachman and TOGAF outputs) to ensure continued alignment with institutional goals, emerging technologies, and new educational standards

The PSU OJT Management and Monitoring Portal serves as a significant step toward PSU's digital transformation. Its implementation will not only simplify OJT coordination but also reinforce the university's commitment to innovation, data-driven governance, and student-centered service delivery. Through continuous improvement and institutional support, the system will evolve into a cornerstone platform for academic modernization across Pangasinan State University.